

Domain Research Note

Financial Question Answering · Hallucination Detection in LLMs · Indian Financial News

Project Domain	Financial NLP — Hallucination Detection in RAG Systems
Dataset	kdave/IndianFinancialNews (HuggingFace) — 26,000 articles
Milestone	Milestone 1 — April 30, 2026
Approach	RAG Pipeline + LLM-as-Judge Validator + SelfCheckGPT Ablation

1. The Problem — Why Financial Hallucination is Costly

Large Language Models (LLMs) like GPT-4 and Llama3 are increasingly deployed in financial services — for earnings summaries, investment research, and regulatory filings. However, LLMs are known to **hallucinate**: generate fluent, confident text that is factually wrong or unsupported by source material. In general domains this is inconvenient; in finance it is **potentially costly**.

Consider: an LLM retrieves an article stating “*RBI raised repo rate to 6.75%*” but generates the answer “*RBI cut rates to support growth.*” A trader or analyst acting on this hallucinated output could make incorrect position decisions. SEBI data shows that automated financial tools increasingly influence retail investor behaviour — making LLM reliability a consumer protection concern, not just a technical one.

Financial hallucination categories (from literature review):

- **Numerical errors** — wrong percentages, figures, dates
- **Attribution errors** — claims not found in source documents
- **Contradiction** — answer conflicts with retrieved context
- **Entity confusion** — wrong company, person, or instrument named

2. Our Approach — RAG with LLM-as-Judge Validation

Standard RAG (Retrieval-Augmented Generation) reduces hallucination by grounding the LLM in retrieved documents. However, it does *not verify* whether the generated answer actually stays within the retrieved content. We add a **second LLM call — the Validator** — that inspects every sentence of the answer and labels it SUPPORTED, UNSUPPORTED, or CONTRADICTED with respect to the source chunks.

Component	Role	Technology
Generator	Find top-3 relevant news chunks	FAISS + all-MiniLM-L6-v2
	Produce grounded answer	Groq / Llama3-8B
Validator	Label each sentence (core contribution)	Groq / Llama3-8B (2nd call)
SelfCheckGPT	Consistency score across 3 samples	BERTScore / Jaccard overlap

3. Research Foundation

The three papers below directly underpin our methodology:

Paper	Key Contribution	Used In
SelfCheckGPT Manakul et al., 2023 arXiv:2303.08896	Sample same query N times; inconsistency = hallucination signal. Zero-resource — no reference corpus needed.	<code>selfcheck.py</code> <i>ablation baseline</i>
Self-RAG Asai et al., 2023 arXiv:2310.11511	LLM learns to retrieve on demand and critique its own output using reflection tokens (ISREL, ISUP, ISUSE).	<i>Inspiration for Validator prompt design</i>
LRP4RAG Anonymous, 2024 arXiv:2408.15533	Layer-wise Relevance Propagation traces which retrieved tokens contributed to each generated token — detects fabrication.	<i>Future work / Week 2 extension</i>

4. Evaluation Metrics

All metrics below will be reported in the Final Evaluation (May 15, 2026):

Metric	Definition	Target
Hallucination Rate	% sentences labeled UNSUPPORTED or CONTRADICTED	< 20%
Support Rate	% sentences labeled SUPPORTED by context	> 80%
Validator Precision	Of flagged hallucinations, how many are truly wrong? (50 human-labeled)	> 0.75
Validator Recall	Of true hallucinations, how many did the validator catch?	> 0.70
SelfCheck Consistency	Pairwise BERTScore across 3 samples (0=random, 1=identical)	Correlate with halluc. rate

5. Project Strength — Why This Is Non-Trivial

The following design decisions make this project substantively challenging beyond a basic RAG implementation:

Challenge	Our Solution
Validator may hallucinate itself (“meta-hallucination”)	Compare validator labels vs. human ground truth on 50 examples (Precision/Recall)
SelfCheckGPT and Validator may disagree	Cases of disagreement = most interesting analysis in the report section
No ground-truth QA pairs for financial news	Create 50 human-annotated examples; measure inter-rater agreement (Cohen’s κ)
One chunk size may miss context	Chunk size ablation: 150 / 250 / 400 words — does retrieval quality change?
Validator prompt design affects results	Prompt ablation: strict vs. lenient vs. chain-of-thought prompts